Exploring Patterns of Environmental Justice

The legacy of redlining in Los Angeles County

Ashwin Chockkalingam

2026-10-24

Summary

In this assignment I explored the legacy of redlining in Los Angeles County and analyzed whether there was correlation between HOLC redlining grades and adverse environmental, socioeconomic, and safety outcomes.

Setup

```
# Load in libraries
library(sf)       # for vector data
```

```
Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(stars)    # for raster data
```

```
Loading required package: abind
```

```
library(tmap)     # for static and interactive maps
library(here)
```

```
here() starts at /home/ashwin_chockkalingam/MEDS/EDS-223-Geospatial/eds223-
hwk2
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2

-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(gt)       # for formatted tables
```

```
Attaching package: 'gt'

The following object is masked from 'package:tmap':

    metro
```

```
# Global option setting
options(tibble.width = Inf)
```

Data import

Three datasets were used in this analysis. EJScreen from the EPA for block groups and information on environmental and socioeconomic outcomes. HOLC data from the University of Richmond, and Bike and Pedestrian Crash Reports from the Statewide Integrated Traffic Records System aka SWITRS.

```
# Load in the data
## Read in HOLC Redlining json.
holc <- st_read(here("data", "mapping-inequality",
                     "mapping-inequality-los-angeles.json"))
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/home/ashwin_chockkalingam/MEDS/EDS-223-Geospatial/eds223-hwk2/data/mapping-
inequality/mapping-inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
## Read in EJ Screen gdb.
ejscreen <- st_read(here("data", "ejscreen",
                         "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/ashwin_chockkalingam/MEDS/EDS-223-Geospatial/eds223-hwk2/data/ejscreen/EJSCREEN_2023
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
## Read in Bike/Pedestrian Crashes
crashes <- read_csv(here("data", "SWITRS_LA", "Crashes.csv"))
```

```
Rows: 251100 Columns: 80
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (45): COLLISION_TIME, OFFICER_ID, REPORTING_DISTRICT, CHP_BEAT_TYPE, CI...
dbl  (33): CASE_ID, ACCIDENT_YEAR, JURIS, DAY_OF_WEEK, CHP_SHIFT, POPULATION...
date  (2): PROC_DATE, COLLISION_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Coordinate reference systems

CRSs need to match before we can do any spatial operations. HOLC is EPSG:4326 (geographic, degrees); EJScreen is EPSG:3857 (projected, meters). HOLC was standardized to EJScreen's CRS EPSG:3857.

```r
# Check CRS of each dataset
st_crs(holc)
```

```
Coordinate Reference System:
  User input: WGS 84
  wkt:
GEOGCRS["WGS 84",
    DATUM["World Geodetic System 1984",
        ELLIPSOID["WGS 84",6378137,298.257223563,
            LENGTHUNIT["metre",1]]],
    PRIMEM["Greenwich",0,
        ANGLEUNIT["degree",0.0174532925199433]],
    CS[ellipsoidal,2],
        AXIS["geodetic latitude (Lat)",north,
            ORDER[1],
            ANGLEUNIT["degree",0.0174532925199433]],
        AXIS["geodetic longitude (Lon)",east,
            ORDER[2],
            ANGLEUNIT["degree",0.0174532925199433]],
    ID["EPSG",4326]]
```

```r
st_crs(ejscreen)
```

```
Coordinate Reference System:
  User input: WGS 84 / Pseudo-Mercator
  wkt:
PROJCRS["WGS 84 / Pseudo-Mercator",
    BASEGEOGCRS["WGS 84",
        ENSEMBLE["World Geodetic System 1984 ensemble",
            MEMBER["World Geodetic System 1984 (Transit)"],
            MEMBER["World Geodetic System 1984 (G730)"],
            MEMBER["World Geodetic System 1984 (G873)"],
            MEMBER["World Geodetic System 1984 (G1150)"],
            MEMBER["World Geodetic System 1984 (G1674)"],
            MEMBER["World Geodetic System 1984 (G1762)"],
            MEMBER["World Geodetic System 1984 (G2139)"],
            MEMBER["World Geodetic System 1984 (G2296)"],
            ELLIPSOID["WGS 84",6378137,298.257223563,
                LENGTHUNIT["metre",1]],
            ENSEMBLEACCURACY[2.0]],
        PRIMEM["Greenwich",0,
```

```
            ANGLEUNIT["degree",0.0174532925199433]],
        ID["EPSG",4326]],
    CONVERSION["Popular Visualisation Pseudo-Mercator",
        METHOD["Popular Visualisation Pseudo Mercator",
            ID["EPSG",1024]],
        PARAMETER["Latitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8801]],
        PARAMETER["Longitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8802]],
        PARAMETER["False easting",0,
            LENGTHUNIT["metre",1],
            ID["EPSG",8806]],
        PARAMETER["False northing",0,
            LENGTHUNIT["metre",1],
            ID["EPSG",8807]]],
    CS[Cartesian,2],
        AXIS["easting (X)",east,
            ORDER[1],
            LENGTHUNIT["metre",1]],
        AXIS["northing (Y)",north,
            ORDER[2],
            LENGTHUNIT["metre",1]],
    USAGE[
        SCOPE["Web mapping and visualisation."],
        AREA["World between 85.06°S and 85.06°N."],
        BBOX[-85.06,-180,85.06,180]],
    ID["EPSG",3857]]
```

```
# crs is not the same, so one has to be transformed before any spatial operation
if (st_crs(holc) != st_crs(ejscreen)) {
  warning("HOLC and EJScreen CRSs do not match. Transforming HOLC to match EJScreen below.")
}
```

```
Warning: HOLC and EJScreen CRSs do not match. Transforming HOLC to match
EJScreen below.
```

Part 1: Legacy of redlining in current environmental (in)justice

Preparing the data

We need to restrict EJScreen to Los Angeles County. LA County was used rather than city because the HOLC data extend past the City of Los Angeles so a city filter would discard HOLC neighborhoods the analysis is meant to cover.

```
# check how many rows or block groups LA County has, as a check on the filter below
expected_la_rows <- sum(ejscreen$CNTY_NAME == "Los Angeles County", na.rm = TRUE)
expected_la_rows
```

```
[1] 6591
```

```
# filter ejscreen for LA County and create new dataframe
ejscreen_la <- ejscreen |>
  filter(CNTY_NAME == "Los Angeles County")

nrow(ejscreen_la)
```

```
[1] 6591
```

```
# transform crs to be the same for both datasets
holc_transform <- st_transform(holc, crs = st_crs(ejscreen_la))

# a mismatch here would silently corrupt every spatial operation downstream
if (st_crs(holc_transform) != st_crs(ejscreen_la)) {
  stop("CRS mismatch persists after transform. Do not proceed to the spatial join.")
}

st_crs(holc_transform) == st_crs(ejscreen_la)
```

```
[1] TRUE
```

```
# check for NA values in grade
n_ungraded <- sum(is.na(holc_transform$grade))
n_ungraded
```

```
[1] 1
```

```r
# 1 na value, so i will drop it. this also means NA grade after the join
# unambiguously means "outside the HOLC footprint"
holc_transform <- holc_transform %>%
  drop_na(grade)

# Confirm only the expected four grades remain
if (!setequal(unique(holc_transform$grade), c("A", "B", "C", "D"))) {
  stop("Unexpected grade values remain after dropping NAs.")
}
```

Task 1: Map of historical redlining neighborhoods

The colors below match the palette used on the original 1939 HOLC residential security maps. Block groups are drawn underneath as a context layer/basemap so the reader can locate the HOLC neighborhoods within the county, and the extent is bounded to the HOLC footprint rather than the full county.

```r
# creating map for part 1
tm_shape(ejscreen_la, bbox = holc_transform) +
  tm_borders() +
tm_shape(holc_transform) +
  tm_fill(
    fill = "grade",
    fill.scale = tm_scale_categorical(
      values = c('#76A865', '#7B9BB0', '#CDB44C', '#BC5F72'),
      labels = c('A (First Grade - Best)',
                 'B (Second Grade - Still Desirable)',
                 'C (Third Grade - Definitely Declining)',
                 'D (Fourth Grade - Hazardous)')
    ),
    fill.legend = tm_legend(title = 'HOLC Grade', position = tm_pos_out("right"))
  ) +
  tm_title('Historical Redlining Neighborhoods in Los Angeles County') +
  tm_scalebar() +
  tm_compass()
```

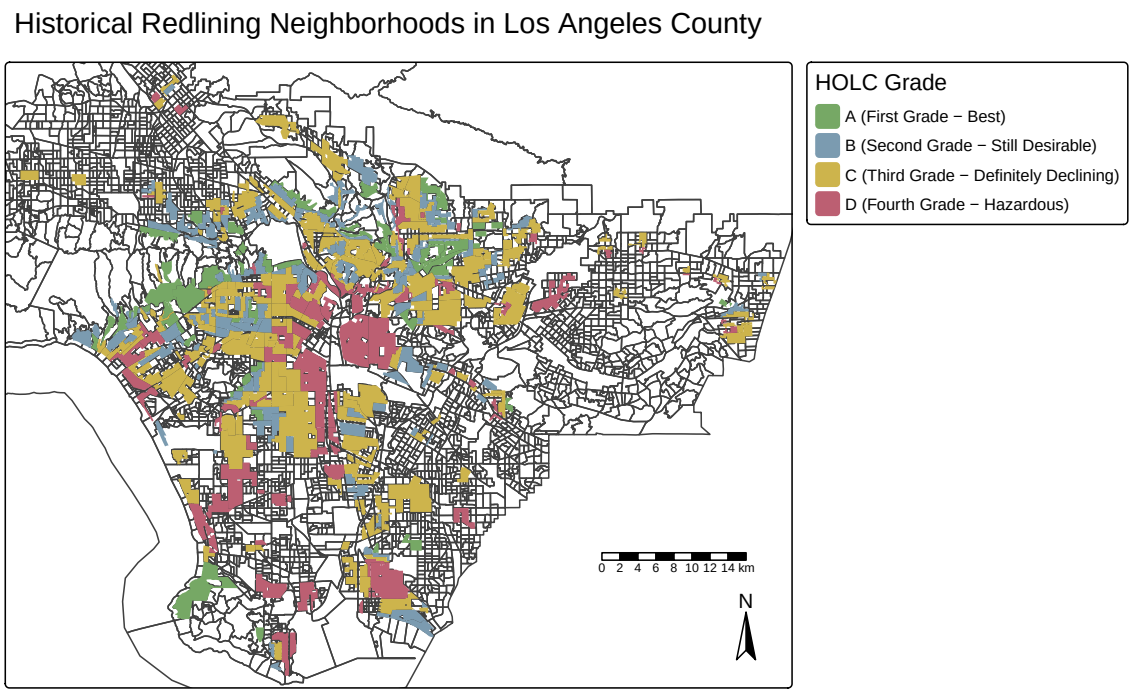


Figure 1: HOLC residential security grades, Los Angeles County

Task 2: Block groups by HOLC grade

```
# Record the pre-join count so duplication can be detected
n_block_groups <- nrow(ejscreen_la)

# Join datasets. largest = TRUE assigns each block group the grade of the HOLC
# polygon it overlaps most which prevents one block group generating multiple
# rows when it straddles a HOLC boundary.
ej_holc_comb <- st_join(ejscreen_la, holc_transform, largest = TRUE)
```

```
Warning: attribute variables are assumed to be spatially constant throughout
all geometries
```

```r
# A duplicated ID would double-count a block group in every percentage below
if (any(duplicated(ej_holc_comb$ID))) {
  stop("Duplicate block group IDs found after the join.")
}

# drop geometry, no longer needed for the summary statistics
ej_holc_comb_df <- st_drop_geometry(ej_holc_comb)

nrow(ej_holc_comb_df)
```

```
[1] 6591
```

The denominator below is all 6,591 LA County block groups. The "No HOLC Grade" row combines areas developed after 1939, and unsurveyed gaps inside the basin.

```r
# create table. count() retains NA as its own category
table_data <- ej_holc_comb_df %>%
  count(grade) %>%
  mutate(percentage = (n / sum(n)) * 100)

# Counts must reconcile to the full set of LA County block groups
if (sum(table_data$n) != n_block_groups) {
  stop("Grade counts do not sum to the total number of block groups.")
}

# outputting table
table_data %>%
  gt() %>%
  tab_header(
    title = "Percentage of census block groups that fall within a HOLC Grade or not.",
    subtitle = "Los Angeles County block groups, n = 6,591"
  ) %>%
  cols_label(
    grade = "HOLC Grade",
    n = "Count of Block Groups",
    percentage = "Percentage"
  ) %>%
  sub_missing(columns = grade, missing_text = "No HOLC Grade") %>%
  fmt_number(columns = percentage, decimals = 2)
```

Table 1

Percentage of census block groups that fall within a HOLC Grade or not.

Los Angeles County block groups, n = 6,591

HOLC Grade	Count of Block Groups	Percentage
A	261	3.96
B	671	10.18
C	2062	31.29
D	958	14.53
No HOLC Grade	2639	40.04

Task 3: Present-day conditions within HOLC grades

The three variables are split across two figures because they are measured on different scales. The two percentile measures share a 0 to 100 scale and can sit on one axis, while % low income is a percentage of households and needs its own. Ungraded block groups are excluded from both so that the comparison is only for the grades (A-D).

```
# create summary df for percentile chart; ungraded block groups excluded
percen_chart <- ej_holc_comb_df %>%
  filter(!is.na(grade)) %>%
  group_by(grade) %>%
  summarize(
    avg_particulate = mean(P_PM25, na.rm = TRUE),
    avg_life_exp    = mean(P_LIFEEXPPCT, na.rm = TRUE)
  )

# pivot data to long format so that measure can be mapped to fill
percen_chart_long <- percen_chart %>%
  pivot_longer(
    cols = c(avg_particulate, avg_life_exp),
    names_to = 'Variable',
    values_to = 'Value'
  )

# create graph
percen_chart_long %>%
  ggplot(aes(x = grade, y = Value, fill = Variable)) +
  geom_col(position = "dodge") +
```

```
  labs(
    title = 'Average percentile for low life expectancy and average \npercentile for particul
    x = 'HOLC Grade',
    y = 'Average Percentile',
    fill = 'Measure'
  ) +
  scale_fill_discrete(
    labels = c(avg_life_exp = "Average Low Life Expectancy",
               avg_particulate = "Average Particulate Matter (2.5)")
  )
```

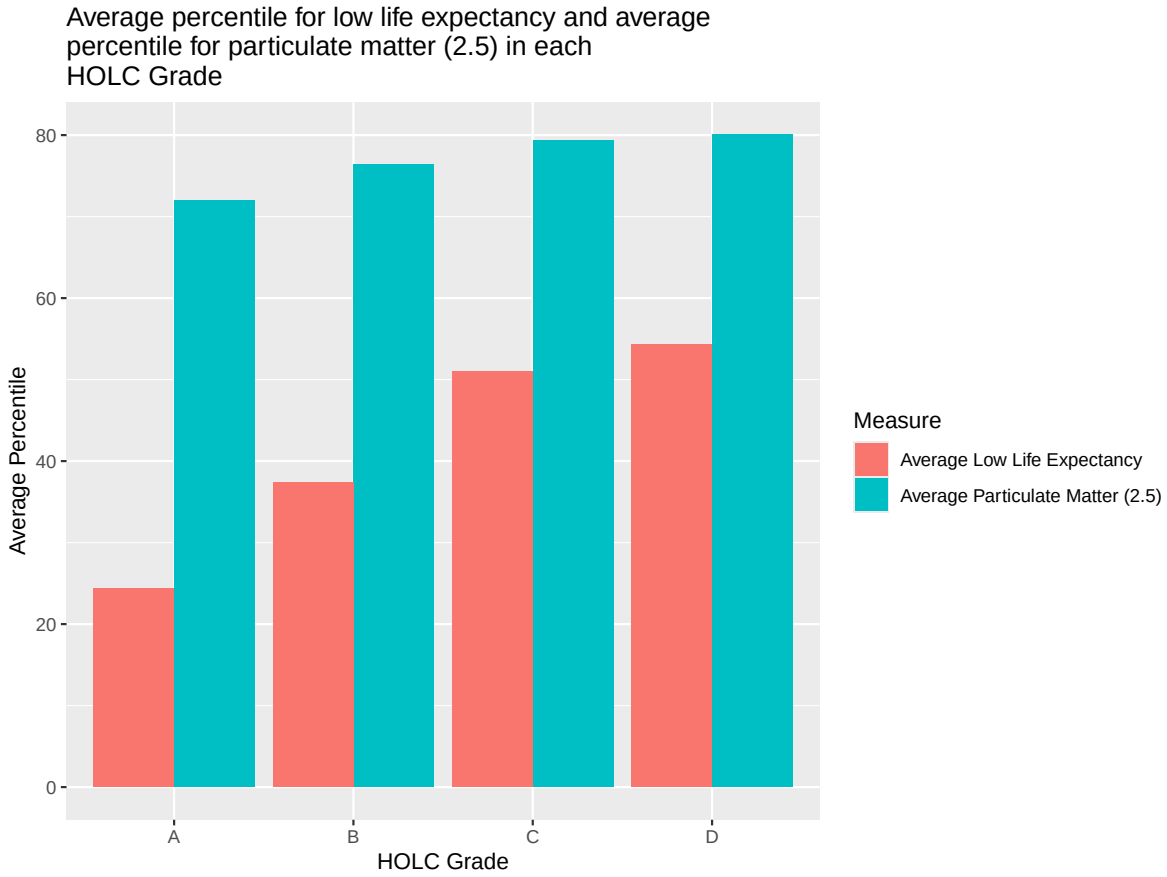


Figure 2: Mean environmental and health percentiles by HOLC grade

```
# low income pct plot. LOWINCPCT is stored as a proportion, so multiply by 100
# to express as a percentage
low_income_plt <- ej_holc_comb_df %>%
```

```
  filter(!is.na(grade)) %>%
  group_by(grade) %>%
  summarize(avg_low_income = 100 * mean(LOWINCPCT, na.rm = TRUE))

# create graph
low_income_plt %>%
  ggplot(aes(x = grade, y = avg_low_income)) +
  geom_col(fill = 'darkred', color = 'darkred') +
  labs(
    title = 'Average Percentage of Low-income Households in \nEach HOLC Grade',
    x = 'HOLC Grade',
    y = 'Average Percentage'
  )
```

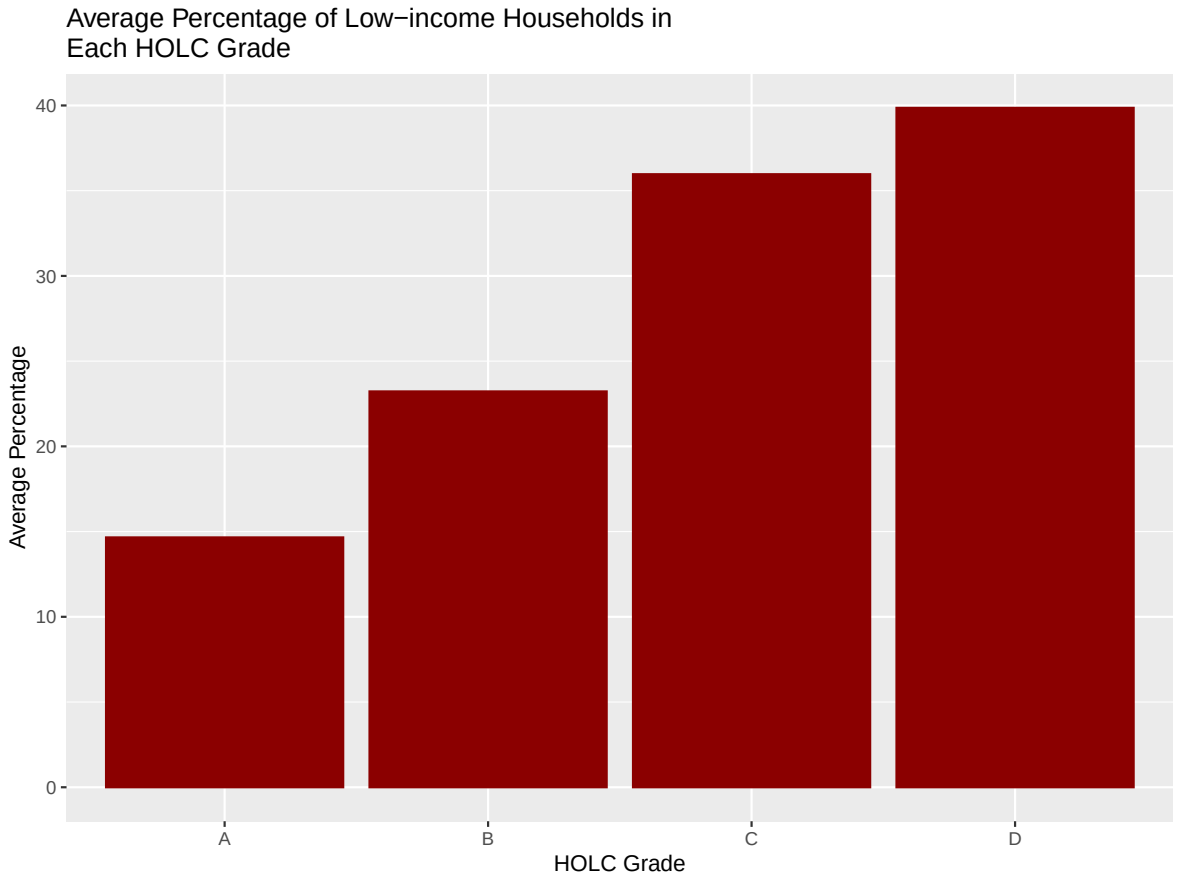


Figure 3: Mean percentage of low-income households by HOLC grade

Task 4: Reflection

All three of the metrics we analyzed (air quality, share of low income, and share of low life expectancy) worsened from grade A (best) to D (worst). The two percentile metrics are different in terms of their steepness with the low life expectancy gradient covering the 24th to 54th percentile whereas the PM2.5 gradient is from the 72nd to 80th (which suggests that air quality is a regional burden across LA whereas poverty and health outcomes correlate with grade more closely) These areas were also different prior to HOLC so it's not necessarily the sole factor in the cause of the disparities but it definitely played a part. HOLC helped in formalising and entrenching the disparities that were already there.

Part 2: Legacy of redlining in pedestrian and bike accidents

Preparing the crash data

SWITRS codes crash types as flag columns taking "Y" or blank, so an NA means the crash was not of that type rather than that data is missing. The filter below keeps crashes flagged as either pedestrian or bicycle.

```
# filtering for bike and pedestrian incidents
nrow(crashes)
```

```
[1] 251100
```

```
# SWITRS flags crash types as "Y" or blank, so NA means "not this type"
bike_ped_crash <- crashes %>%
  filter(PEDESTRIAN_ACCIDENT == "Y" | BICYCLE_ACCIDENT == "Y")

if (nrow(bike_ped_crash) == 0) {
  stop("No bike or pedestrian crashes matched. Check the flag column coding.")
}

nrow(bike_ped_crash)
```

```
[1] 34733
```

The codebook documents POINT_X/POINT_Y as WGS84 coordinates geocoded by SafeTREC, while LATITUDE/LONGITUDE come straight from the crash report with no stated CRS. POINT_X/POINT_Y are also far more complete, as the counts below show.

```
# count NAs for two types of location information
n_na_point <- sum(is.na(bike_ped_crash$POINT_X))
n_na_latlon <- sum(is.na(bike_ped_crash$LATITUDE))

n_na_point
```

```
[1] 1634
```

```
sum(is.na(bike_ped_crash$POINT_Y))
```

```
[1] 1634
```

```
sum(is.na(bike_ped_crash$LONGITUDE))
```

```
[1] 21139
```

```
n_na_latlon
```

```
[1] 21139
```

Checking the bounding box after conversion revealed records sitting at exactly (0, 0), meaning some rows use 0 as a missing-coordinate placeholder rather than NA, plus a small number of geocodes falling outside the county. Filtering spatially to LA County removes both.

```
# going to use point x/y because long/lat is missing way more.
# filter out na values in points, then declare the CRS the coordinates are
# already in (WGS84 per codebook) and transform to match the other datasets
bike_ped_crash_sf <- bike_ped_crash %>%
  filter(!is.na(POINT_X)) %>%
  st_as_sf(coords = c("POINT_X", "POINT_Y"), crs = 4326) %>%
  st_transform(crs = st_crs(holc_transform))

# Verify the conversion landed in Los Angeles rather than somewhere else
st_bbox(bike_ped_crash_sf)
```

```
     xmin      ymin      xmax      ymax
-13569664         0         0   5089995
```

```
if (st_crs(bike_ped_crash_sf) != st_crs(holc_transform)) {
  stop("Crash CRS does not match HOLC after transform.")
}

n_before_filter <- nrow(bike_ped_crash_sf)

# Restrict to crashes falling inside LA County. Removing records that used
# 0 as a missing-coordinate placeholder.
bike_ped_crash_sf <- bike_ped_crash_sf %>%
  st_filter(ejscreen_la)

n_dropped_spatial <- n_before_filter - nrow(bike_ped_crash_sf)

if (n_dropped_spatial > 0) {
  warning(paste("Dropped", n_dropped_spatial,
                "crash records falling outside Los Angeles County",
                "(placeholder (0,0) coordinates and erroneous geocodes)"))
}
```

```
Warning: Dropped 101 crash records falling outside Los Angeles County
(placeholder (0,0) coordinates and erroneous geocodes)
```

```
# Confirming the extent is now in LA County
st_bbox(bike_ped_crash_sf)
```

```
     xmin      ymin      xmax      ymax
-13240187   3940420 -13099645   4139443
```

```
nrow(bike_ped_crash_sf)
```

```
[1] 32998
```

Joining crashes to HOLC grades

Roughly 52% of crashes fall outside the HOLC data. The figure below reports each measure as a share of crashes only within graded areas.

```r
n_crashes <- nrow(bike_ped_crash_sf)

# spatial join. Points fall within a single polygon, so the default is sufficient
crash_holc <- st_join(bike_ped_crash_sf, holc_transform)

# Crashes outside the HOLC footprint, excluded from the figure below
n_ungraded_crashes <- sum(is.na(crash_holc$grade))
n_ungraded_crashes
```

```
[1] 17298
```

```r
# drop geometry
crash_holc_df <- st_drop_geometry(crash_holc)
```

```r
# Count crashes and sum fatalities within each grade. The two fatality columns
# are summed separately so that an NA in one does not discard the other.
crash_summary <- crash_holc_df %>%
  filter(!is.na(grade)) %>%
  group_by(grade) %>%
  summarise(
    crashes = n(),
    fatalities = sum(COUNT_PED_KILLED, na.rm = TRUE) +
                 sum(COUNT_BICYCLIST_KILLED, na.rm = TRUE)
  ) %>%
  mutate(
    pct_crashes    = crashes / sum(crashes) * 100,
    pct_fatalities = fatalities / sum(fatalities) * 100
  )

# Fatalities cannot exceed crashes within any grade
if (any(crash_summary$fatalities > crash_summary$crashes)) {
  stop("Fatality count exceeds crash count in at least one grade.")
}

crash_summary
```

```
# A tibble: 4 x 5
  grade crashes fatalities pct_crashes pct_fatalities
  <chr>   <int>      <dbl>       <dbl>          <dbl>
1 A         360          7        2.29          0.923
2 B        1782         76       11.4          10.0
```

```
3 C        9026        439       57.5          57.9
4 D        4532        236       28.9          31.1
```

```r
# pivot to long format so that measure can be mapped to fill
crash_summary_long <- crash_summary %>%
  select(grade, pct_crashes, pct_fatalities) %>%
  pivot_longer(
    cols = c(pct_crashes, pct_fatalities),
    names_to = 'Measure',
    values_to = 'Percentage'
  )

# create graph
crash_summary_long %>%
  ggplot(aes(x = grade, y = Percentage, fill = Measure)) +
  geom_col(position = "dodge") +
  labs(
    title = 'Distribution of bike and pedestrian crashes and fatalities \nacross HOLC grades
    subtitle = 'Each measure shown as a share of its own total within HOLC-graded areas',
    x = 'HOLC Grade',
    y = 'Percentage (%)',
    fill = 'Measure'
  ) +
  scale_fill_discrete(
    labels = c(pct_crashes = "Crashes", pct_fatalities = "Fatalities")
  )
```

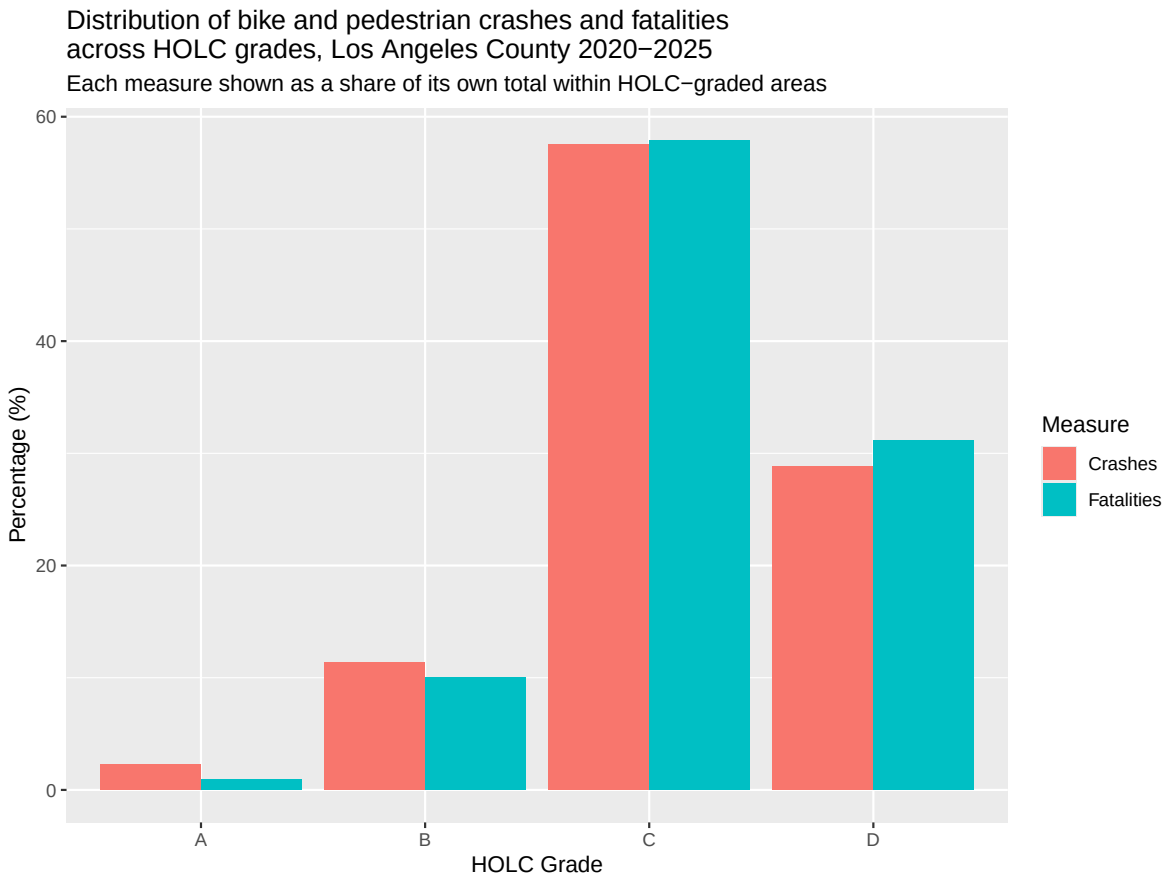


Figure 4: Distribution of bike and pedestrian crashes and fatalities across HOLC grades

The two distributions are very similar. With this data we cant say with confidence that a crash involving a bike or pedestrian in a C or D area is more likely to be fatal than one in any other area, it's just that there are more records of them possibly due to there being more people.

Data sources

EJScreen. United States Environmental Protection Agency. *EJScreen: Environmental Justice Screening and Mapping Tool*, 2023 release, block group level with state percentiles. The tool has since been decommissioned; data downloaded from https://www.epa.gov/ejscreen/download-ejscreen-data.

HOLC redlining grades. Robert K. Nelson, LaDale Winling, Richard Marciano, Nathan Connolly, et al. "Mapping Inequality: Redlining in New Deal America." *American Panorama*,

ed. Robert K. Nelson and Edward L. Ayers. Digital Scholarship Lab, University of Richmond. https://dsl.richmond.edu/panorama/redlining/

Crash records. California Highway Patrol. *Statewide Integrated Traffic Records System (SWITRS)*, 2020-2025. Accessed via the Transportation Injury Mapping System (TIMS), Safe Transportation Research and Education Center, University of California, Berkeley. https://tims.berkeley.edu/